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Experimental calibration of flat jacks for in-situ testing of masonry

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ABSTRACT

Flat-jack testing method is one of the most commonly used techniques for the structural assessment of existing masonry structures. Single and double flat jacks are usually adopted to evaluate the acting normal stress, or the compressive behaviour of masonry material. Test procedures are codified by international standards (e.g., A.S.T.M D4729-87; C1196-04; C1197-04, R.I.L.E.M TC 177-MDT D.4; R.I.L.E.M. TC 177-MDT D.5), which provide the preliminary calibration of an experimental coefficient (k_m) and of the effective area (A_{eff}), which determination influences significantly the reliability of the test. This article presents the result of an experimental study on the calibration of flat jacks for masonry testing. The problem is investigated by several tests carried out on two types of common flat jacks, which differ for geometry and producer. Two calibration methods are adopted in order to relate the pressure values of the flat jack with those of the hydraulic press and load cycles are performed in two different pressure ranges. Finally, a theoretical interpretation of results is made, which gives good predictions of calibration parameters. Results of this investigation highlight the influence of constructive features and service pressure of the jack adopted, for obtaining reliable results from the tests on masonry structures.

KEYWORDS

compressive stress;
conversion factor; flat-jack
method; masonry

1. Introduction

Flat-jack testing method is one of the most commonly used techniques for structural investigations of existing masonry structures. The method was originally proposed in the field of geotechnical engineering, where it is still applied, especially in the form of overcoring testing (Campione and Minafò 2014), for assessing the stress state in rock masses. The proved efficiency of this method led to wide applications in masonry structures since the 1980s (Rossi 1987), and nowadays it represents one of the most popular techniques for in situ testing.

As it is well-known from engineering practice (Binda and Tiraboschi 1999; Gregorczyk and Lourenco 2000), the flat-jack method can be performed in two ways: single or double flat jacks, on the basis of the goal to be achieved. The single flat-jack method aims to evaluate the acting normal stress, while the double flat jacks to evaluate the mechanical parameters of masonry material, with particular reference to the elasticity modulus in compression. Single flat-jack testing allows to measure normal stress in a small area of the structure by a plane cut, which is made in general in correspondence of a mortar bed joint when masonry is quite regular (e.g., brick and block masonry). The flat jack is then

inserted in the cut and then loaded with a gradually increasing pressure, the latter applied by a hydraulic pump until displacements are recovered to the existing condition before the cut. The necessary pressure to restore the initial distance between the two cutting edges allows obtaining the measure of the acting normal stress. When some information is needed on the compressive behavior of masonry, a second cut is made parallel to the first one and a second jack is inserted. Therefore, the portion of masonry between the two cuts is loaded by simulating a uniaxial compressive stress state. The pressure is applied with subsequent increments until the point of nonlinearity is highlighted in the load-strain curve to avoid damage in the structure. Sometime, it is possible to go beyond this load level, when it is possible to destroy the masonry portion between the two jacks, allowing to obtain the in-situ nonlinear response up to failure. In both cases, technical specifications of the flat jack are needed to obtain the stress measure from the applied pressure. In particular, the recorded pressure should be multiplied for two coefficients: the calibration coefficient of the flat jack, which should be determined in laboratory, and the ratio between the area of the flat jack and the area of

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the cut. The measured stress is finally provided by the following expression:

$$\sigma = k_m \cdot k_a \cdot p, \quad (1)$$

where k_m is the conversion factor, and it depends on the stiffness and shape of the jack adopted, k_a is equal to the ratio between the area of the flat jack and the area of the cut, and p is the pressure applied by the jack.

It is worth noting that this method is valid generally for brick or regular stone masonry, while the case of rubble stone masonry is different. In fact, pieces of stones or bricks can be included in the cut due to the marked heterogeneity of this kind of masonry. As a consequence the application of flat-jack method on rubble stone masonry requires a careful calibration. Some recent applications can be found in Andreini et al. (2014) and Simões et al. (2016).

The reliability of the obtained results depends on the quality of the tests, especially in presence of weak masonry or irregular masonry, in the cases of low values of compressive stresses (low buildings), in the cases of multi-faceted masonry or masonry walls subjected to eccentric loads, or in the presence of consolidation, and so on. The test procedure is codified in American Code A.S.T.M., in European Code R.I.L.E.M.

It is evident as the reliability of the test is strongly influenced by the calibration factors k_m and k_a , which depend on of the geometric and constructive features of the flat jack. A preliminary experimental calibration of the adopted jack should be of fundamental importance before performing a test but however, often it is not a viable way for quick practical applications. Usually, producers provide suggested values for the calibration factor k_m , and constant values are provided, independently by the service pressure.

Due to its usefulness in practical engineering cases, several research works and applications of flat-jack method were presented in the literature. Rossi (1982, 1987) performed the first studies on this subject, presenting the technique and showing its potential when applied to actual case studies. The researcher stated that k_m is a jack constant that must be determined by means of laboratory testing, and performed calibration tests on large-scale samples, by comparing the stress levels induced in the walls with those recorded by the jacks. He concluded that the conversion factor is a function of the geometry of the flat jack but it depends by the pressure level, and the correlation was presented in form of calibration curves. From a theoretical point of view, Sacchi Landriani and Taliercio (1986) verified the reliability of both single and double flat-jack technique by bidimensional numerical simulations.

Gregorczyk and Lourenco (2000) presented an overview on the technique and proposed some additional recommendations to ASTM and RILEM standards, by investigating on the effective contact area of the flat jack. The researchers performed tests on scaled walls and measured experimentally the contact area by varying the technique adopted for making the cut.

Carpinteri, Invernizzi, and Lacidogna (2009) proposed some modifications to the double flat-jack method. In particular, researchers evaluated the adoption of vertical cuts along the masonry portion to be tested, in order to ensure a uniaxial stress state in the masonry prisms. Moreover, they performed tests in combination with acoustic emission sensors, which were able to detect the extension of damage.

Uranjek et al. (2012) carried out a complete in situ diagnostic campaign on a multi-leaf stone masonry building using different techniques. In particular, researchers adopted double flat-jack technique for assessing the efficacy of cement grout injections in enhancing the compressive elastic modulus and strength of stone masonry walls, obtaining good results.

Lombillo et al. (2013) presented an experimental study comparing the results of non-destructive and minor-destructive testing applied in a rubble stone masonry construction. Researchers adopted flat jacks, hole-drilling, and sonic tests for evaluating the mechanical features of different masonry walls, finding good correspondence between the results of each technique.

More recently, Łątka and Matysek (2017) performed laboratory tests by adopting three different shapes of flat jack and various methods for recording displacements during the tests (i.e., Digital Image Correlation). Authors investigated on the precision of mortar joint cutting, analyzing its effect on the effective working area of the jack. Moreover, the comparison between induced stress and corresponding measured value allowed finding the conversion factor for the jacks adopted and comparing their values with those suggested by the producer.

It is evident as despite this method is well known and used in practical cases, some points need to be clarified on the calibration of the flat jack, with particular reference to the parameters affecting the conversion factor and the effective contact area.

This article presents the results of an experimental investigation on the calibration of flat jacks for masonry testing. The problem is investigated by different tests, by using two types of semicircular flat jacks made by different manufacturers, and commonly adopted for practical engineering cases. Two calibration methods are followed based on available international standards, in order to relate the jack pressure values with those of the hydraulic press and aiming to obtain

the conversion coefficient k_m and effective area of the jacks A_{eff} .

Finally, a theoretical interpretation of results is made by an analytical formulation supported by finite element (FE) analyses, aiming to obtain general expressions for calculating k_m and A_{eff} . Results of this investigation highlighted as several parameters affect the calibration of flat jacks, and in particular the constructive features of flat jacks, including their shape and stiffness. Moreover, it is stressed as the reliability of the method is dependent by the pressure level in the jack, and consequently it depends on the stress state and mechanical features of the investigated masonry.

2. Experimental program

The current experimental study provides the calibration of two kinds of flat jacks, commonly adopted for in-situ testing. The target is to assess the coefficient k_m by relating the pressure induced in the jack and the external force recorded by a load cell in a testing machine. The first type of jack, namely B series (Figure 1a), has semi-circular shape, sizes $350 \times 260 \times 4$ mm and it is made with steel foils

having thickness equal to 0.8 mm. The nominal area of this series of jack is equal to $77,506 \text{ mm}^2$. The second type of flat jack, C series (Figure 1b) has a similar shape, but sizes are $345 \times 260 \times 6$ mm, and it is made with 2-mm thick foils of steel. Nominal area of the jack is in this case equal to $76,045 \text{ mm}^2$. For each kind of flat jack, three specimens are considered, each one made by the same producer, for an overall number of six jacks.

2.1 Test set-up and instruments

Test set-up is shown schematically in Figure 2. A force-controlled testing machine with spherical swivels is adopted. The flat jack is introduced between the plates of the machine, which relative distance is kept constant. Three Linear Voltage Displacement Transducers (LVDTs) are arranged to monitor the displacements during the test, two on the sides and one in the front of the flat jack, allowing to check the uniformity of displacements (Figure 2). A load cell is also introduced between a swivel and the steel plate to check the load level. Oil is pumped inside the flat

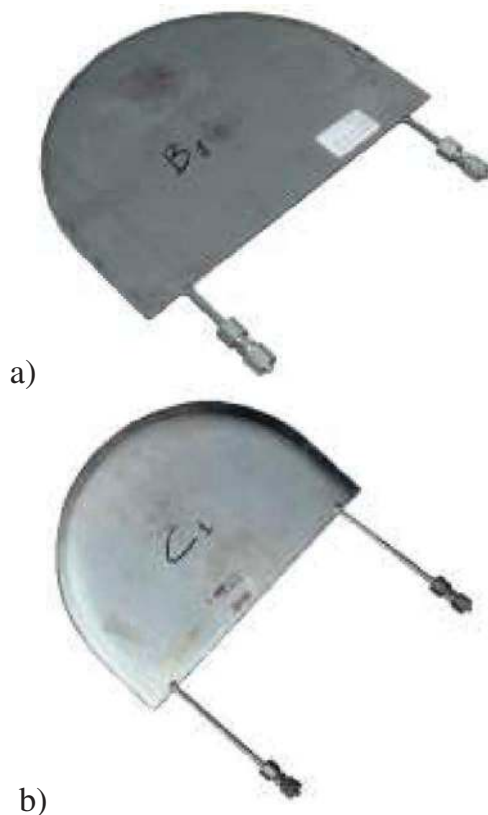


Figure 1. Types of tested flat jacks. (a) Jack B1; Jack C1.

1: spherical swivel and load cell; 2: upper-bearing plate of the testing machine; 3: LVDT; 4: shim; 5: flat jack; 6: bottom-bearing plate of the testing machine; 7: hydraulic line.

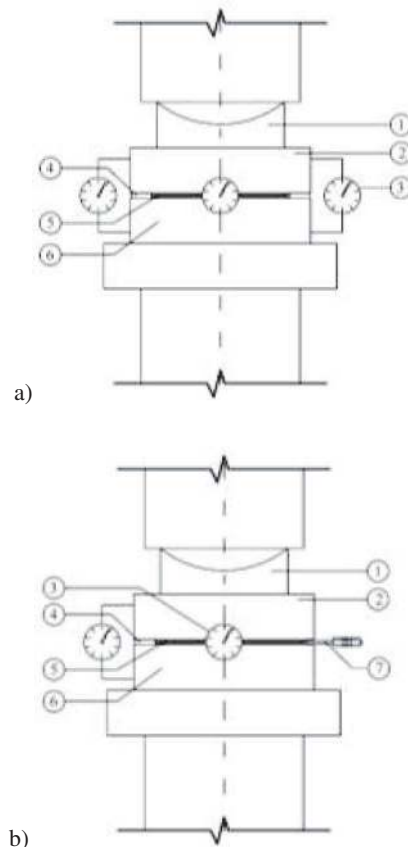


Figure 2. Test set-up: (a) front view; (b) lateral view.

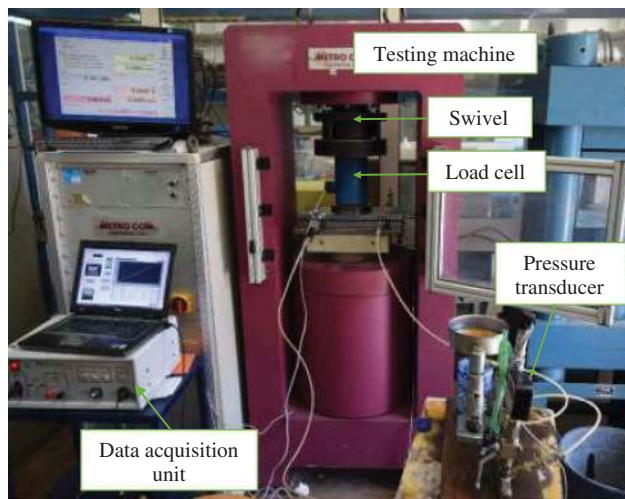


Figure 3. Test on a flat jack.

jack by a hydraulic pump, connected to two digital pressure transducers. LVDTs, load cell, and pressure transducer are connected to a data acquisition unit, which allows storing data that can be managed by a PC, as shown by Figure 3.

2.2 Testing procedure

Both procedures provided by ASTM and RILEM are followed to find the k_m coefficient and effective area A_{eff} . Both technical codes provide to perform three load cycles with constant increases of pressure up to a maximum value of pressure is reached equal to 95% of the maximum service load prescribed by the producer. In particular, in the present study a value of 60 bar is adopted as maximum value for both series of jacks. Each specimen is loaded making difference between low and high pressures. In the first case, the pressure is kept between 2 and 6 bar with increments equal to 1 bar, while in the second range, pressure is maintained between 12 and 60 bar with increments equal to 6 bar. Test is performed by inducing an established value of pressure in the flat jack. Afterward, load is applied from the testing machine in order to keep the distance between the plates equal to 1.33 times the thickness of the jack. In this way, the jack is working as in inverse manner to his common service conditions. At each step, the value of pressure in the transducer is recorded and related to the corresponding load value measured by the load cell. The ratio between load and pressure values represents the effective area of the flat jack A_{eff} , according to the RILEM procedure. Similarly, the ratio between the pressure value multiplied by the nominal area of the jack and the load given by the

load cell represents the k_m coefficient, according to the procedure provided in the ASTM code. In the following, results are presented in terms of RILEM or ASTM representation, and they are not all reported for the sake of brevity.

3. Results

3.1 Load cycles

Figures 4 and 5 show some results in graphical form for specimens of B series. In particular, graphs of Figure 4a, b plot the load recorded by the testing machine as a function of the corresponding load measured by the pressure transducer in the jack, for low and high pressure, respectively. It should be reminded that the slope of this curve represents the k_m coefficient, according to the ASTM procedure, and an ideal jack—only theoretical case—should show a linear trend with slope equal to 45°. The three load cycles are shown in both cases, highlighting as the behaviour is substantially different on the basis of the pressure range. In fact, the trend of the function represented seems to be almost linear for high pressures, with points almost coincident for the three load cycles. Differently, load values recorded for the three cycles are slightly different for low pressures,

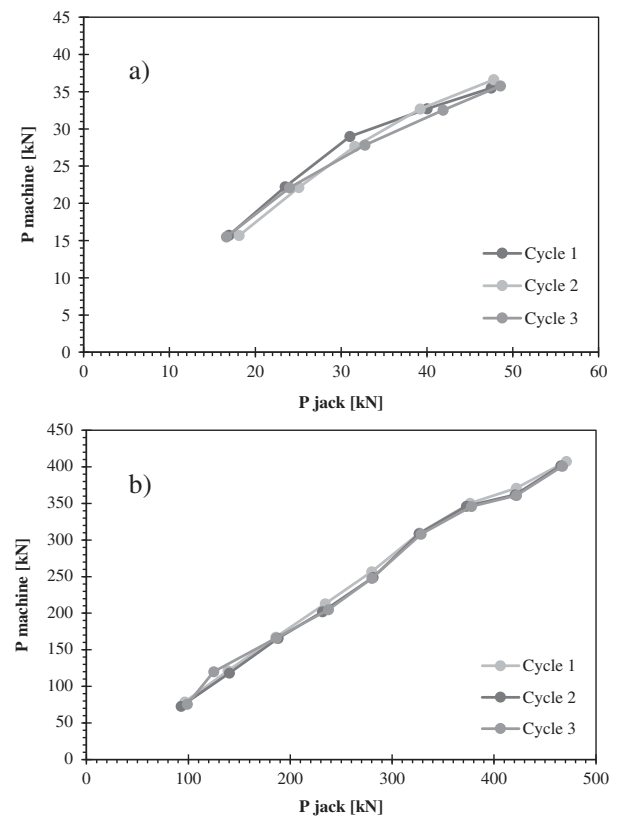


Figure 4. Results for specimen B1: (a) low pressure; (b) high pressure.

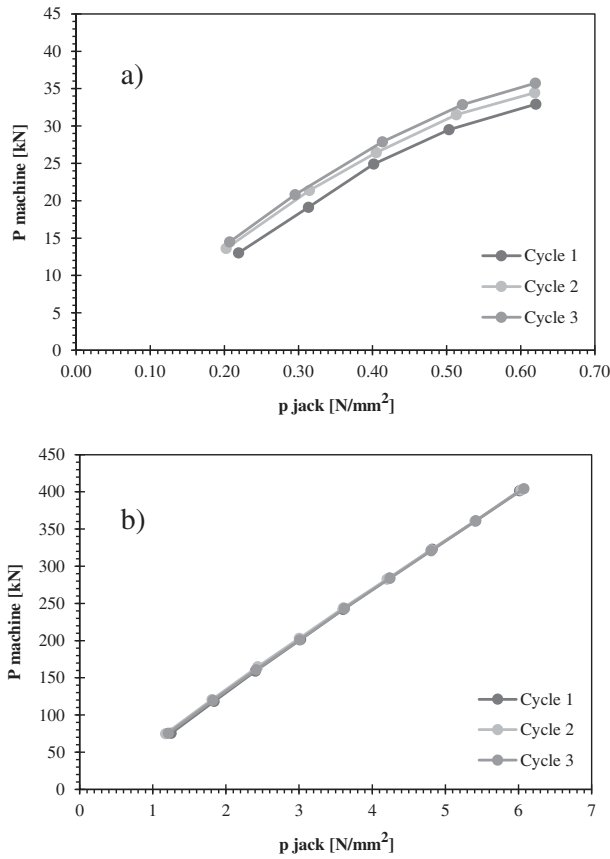


Figure 5. Results for specimen B2: (a) low pressure; (b) high pressure.

and the load applied by the machine tends to be a nonlinear function of the pressure recorded in the jack. Similar results are achieved for specimens B2 and B3. In particular, Figure 5 shows the load recorded by the testing machine as a function of the pressure read by the transducer, for specimen B2. It is worth noting that the slope in these curves represents the effective area of the jack, according to the RILEM code. Also in this case, load cycles for low values of pressure (Figure 5a) are not coincident, and a nonlinear trend is recorded again, with increasing values of load for successive cycles. A marked linear trend is achieved for high pressures (Figure 5b), meaning that the effective area of the jack can be considered constant and only a function of the range of working pressure and of the quality of the cut (Gregorczyk and Lourenco 2000).

Results for flat jack B3 are similar to those obtained for specimens B1 and B2 and they are not shown here for the sake of brevity. Specimens of C-series shows a similar behaviour, but with quite different slopes of recorded curves. As an example, Figure 6 shows data obtained for specimen C1. It is evident as a larger scatter between the three load cycles is achieved for low pressure values and data recorded assumes an

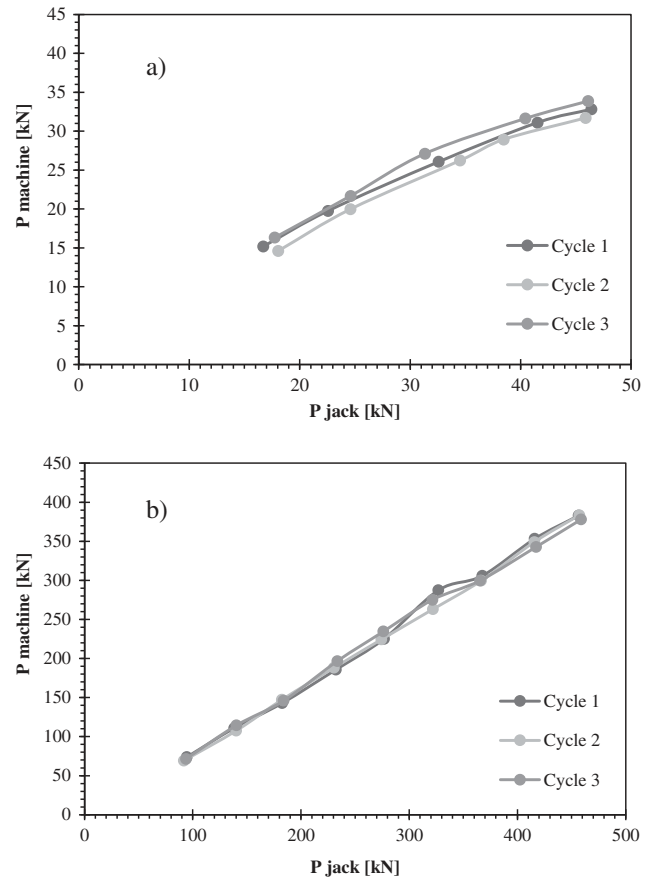


Figure 6. Results for specimen C1: (a) low pressure; (b) high pressure.

irregular trend, particularly for specimen C3. Moreover, it is observed as the slope of curves shown in Figure 6 is substantially lower than 45° , expecting a low value of k_m . Results for specimen C2 and C3 are similar to those obtained for specimen of C1 and they are not here reported for the sake of conciseness.

3.2 Conversion factor and effective area for tested flat jacks

Results shown in the previous section allow calculating the k_m coefficient as the slope of the curve representing the relationship between the load measured by the machine and the pressure induced in the jack multiplied for the nominal area. This calculation is made for each specimen tested and for each load cycle, making difference between low and high pressure, as described above. In particular, the method of ordinary least squares is applied for fitting experimental data obtained for each cycle and a value of k_m is determined as the angular coefficient of the regression line obtained for each single load cycle. Finally, the k_m coefficient is evaluated for each specimen as the average value between the three cycles.

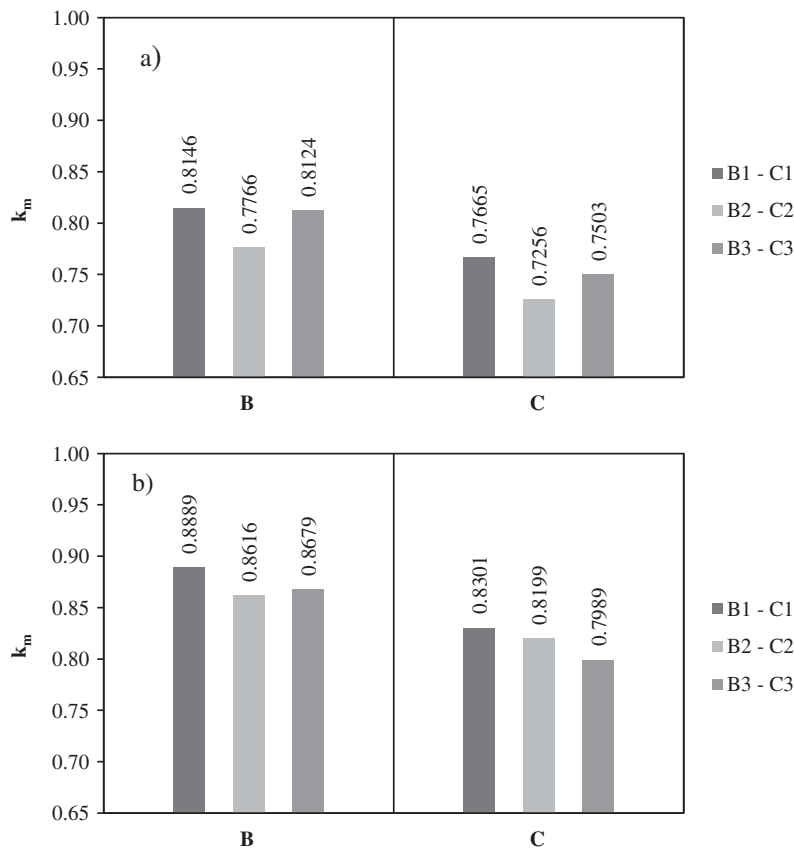


Figure 7. Comparisons of k_m coefficient for the two series: (a) low pressure; (b) high pressure.

Figure 7 shows the comparison between the values of k_m calculated for each specimen as the average value between the three cycles. It is worth noting as lower values are associated to specimens of C-series, which are the stiffer kind of flat-jack. Moreover, larger dispersion of data is observed for C-series specimens. In all cases, the coefficient of the jack is greater for high values of pressure, highlighting as the reliability and the accuracy of the measure performed by flat-jack technique depend from the stress state in the observed masonry wall. It is also worth noting as data obtained for B-series are in accordance with those obtained by Łatka and Matysek (2017), who calibrated the same kind of flat jack. In particular, they suggested $k_m = 0.85$ for $p < 10$ bar and $k_m = 0.86$ for $p > 10$ bar, while average values determined from the present investigations are $k_m = 0.80$ for $2 < p < 6$ bar and $k_m = 0.87$ for $12 < p < 60$ bar.

It is evident as, series C of specimens is characterized by lower values of k_m and larger scattering of results, especially for low pressure values. This fact can be explained considering that for this low range of pressure, oil pumped is not able to fulfil the inner core of the flat jack and large differences are observed.

It is stressed as on the basis of results shown, the adoption of an average experimentally calibrated value, calculated as discussed above, could lead to errors lower than 4% for low pressure and 3% for an higher stress values, which are the scattering values of determined values. This fact highlights that reliability of the test depends by the value of in-situ stress and by constructive features of the jack. It should be also reminded that errors can be achieved on the evaluation of the effective area. This fact is confirmed by results achieved in terms of effective area A_{eff} , the latter obtained as suggested in the RILEM code. In this way, the effective area is deduced as the slope of the regression line drawn to fit the relationship between the load deduced by the machine and the pressure recorded in the jack. Data obtained for each specimen are resumed in Figure 8, which allows a graphical comparison between recorded values. It is confirmed that the two types of flat jack performed differently and higher values are obtained for series B. The trend of values recorded for A_{eff} follows that recorded for k_m , achieving greater values of effective area when the pressure increases. As recorded for k_m , specimens B2 and C2 have shown slightly lower values of A_{eff} with respect to other specimens belonging to the same series, even if maximum

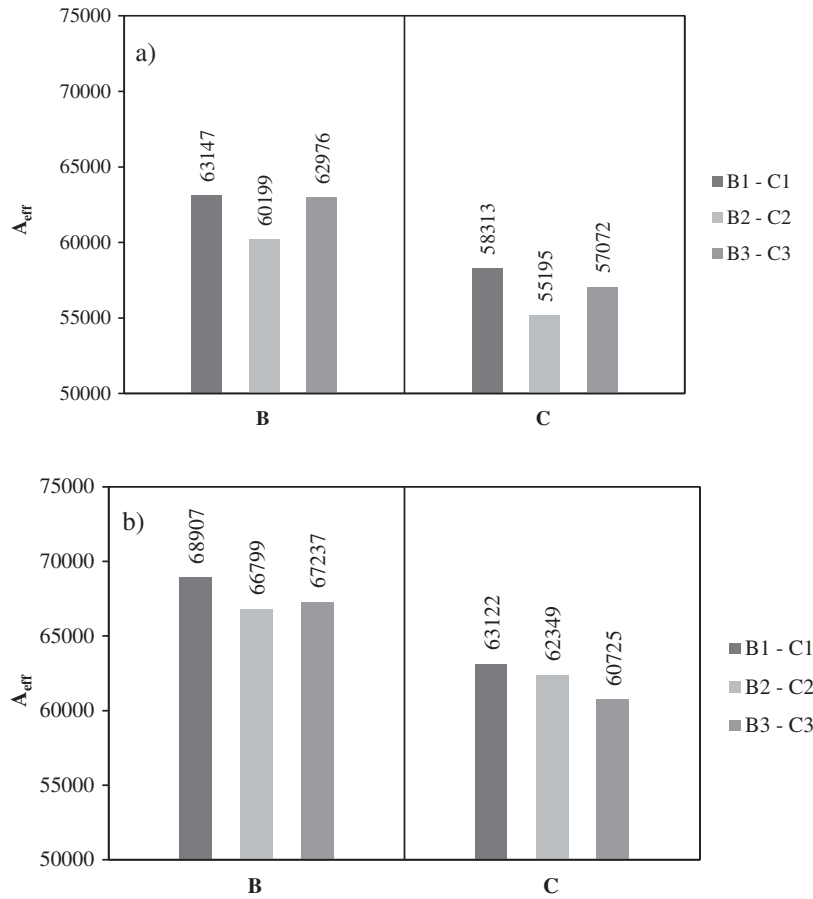


Figure 8. Comparisons of A_{eff} for the two series: (a) low pressure; (b) high pressure.

dispersion is lower than 5.5%. The similarity between the trend of k_m and A_{eff} stresses the fact that these two constants are strictly related each other. However, this consideration is valid due to the fact that their values were deduced by laboratory tests, i.e., deducing the relationship between induced load and recorded pressure with a testing machine. In real applications, it was demonstrated in the literature (Gregorczyk and Lourenco 2000) that the nature of masonry and quality of the cut significantly influence the values of A_{eff} , being the latter greater for brick and block masonry, while it is lower for rubble stone or irregular masonry.

4. Theoretical investigation

It is reminded that flat jacks are made with thin foils of steel, and they are defined as thin envelope-like bladder (ASTM C1196). It is also well known that thin-walled plates can be affected by initial imperfections, due to the manufacturing process. As an example, plate rolling machines despite their objective of creating flat plates, cause surface waves and even overall bows in the surface plus non-uniform thickness. Various shape of initial imperfection can be considered and assumed.

The most simplified and commonly measured is the bow-deformed shape. The maximum amplitude of the initial imperfection can be assumed in the range of $L/400$ and $L/800$ as commonly done for steel members, where L is the maximum in-plane dimension of the jack. In this way, a small initial pressure p_0 in the flat jack is needed to restore the planarity of the plates and warrant the contact with bounding masonry. On the basis of the definition of the k_m factor (Łątka and Matyszek 2017), it can be written in the following form:

$$k_m = \frac{P_{machine}}{p_j \cdot A_{gross}}, \quad (2)$$

where $P_{machine}$ is the load read by the testing machine, p_j is the pressure induced in the jack, and A_{gross} is the nominal area of the flat jack. On the basis of the previous consideration, k_m can be also written as

$$k_m = \frac{(p_j - p_0) \cdot A_{eff}}{p_j \cdot A_{gross}}, \quad (3)$$

If an ideal jack is considered, the effective area can be considered equal to the gross area and the following simplification holds:

$$k_m = 1 - \frac{p_0}{p_j}, \quad (4)$$

Equation (4) expresses the condition that the conversion factor is a function of the working pressure p_j and of its stiffness properties. However, it should be considered that stiffness is not only connected to the necessity of restoring the planarity of the plates, because once of the planarity is restored, part of the internal pressure pumped into the flat jack is used in deforming the plates. Obviously, this pressure used in deforming the plates affects the results of the test much more at low pressures than at high pressures. The proposed approach is based on the assumption that after the planarity is restored the jack is deformable and hence this effect is neglected.

It should be observed that Equation (4) provides $k_m = 1$ for $p_0 = 0$ —ideal jack with no initial imperfection—and also for p_j going to ∞ . The evaluation of p_0 requires the assumption of an initial imperfection shape and amplitude. If a maximum deflection of the initial shape δ_0 is assumed in the centre gravity of a plate, p_0 can be calculated in the following form:

$$p_0 = K_j \cdot \delta_0, \quad (5)$$

where K_j is the stiffness of the jack, meaning the uniform pressure to be applied in the jack to obtain a unit deflection in a considered point. Equations (4) and (5) reflect the condition that the conversion factor decreases for stiffer jacks and for greater initial imperfections.

Aiming to find K_j , tested flat jacks were modelled in the FE code SAP2000 (Computers and Structures, Inc. 2016). In particular, plates were modeled with “Shell Thin” elements, which follows Kirchhoff formulation, neglecting transverse shear deformation. A uniform unit pressure was applied and K_j was found by making the inverse ratio of the obtained maximum deflection. For the sake of clarity, Figure 9 shows the plane view of the FE model built for flat jacks of B-series and the color-plot of vertical displacements under uniform pressure. As expected, the deflection of the jack is not uniform, as it is influenced by its shape and stiffness, being displacements more pronounced in the geometrical centre of the jack’s plate.

The value of p_0 is calculated by Equation (5), after assuming an initial imperfection $\delta_0 = L/400$, obtaining $p_0 = 0.0025$ MPa for B-series and $p_0 = 0.004$ MPa for C-series flat jack, while corresponding values of K_j were equal to 0.00286 MPa/mm and 0.00503 MPa/mm.

Finally, the trend of k_m as a function of the pressure is obtained by considering a coefficient that takes into account the difference between the gross and the net

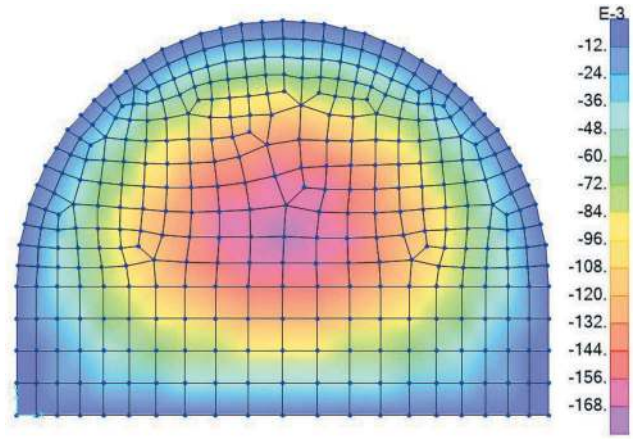


Figure 9. FE model of the flat jack and colour-plot of the deformed shape (units in mm).

area of the jack. The conversion factor is therefore expressed by the following function:

$$k_m = 1 - \left(\frac{p_0}{p_j} \right)^\eta, \quad (6)$$

where η is a coefficient which consider that effective area is different from the gross area. It depends from the quality of the cut and it can be easily estimated by observing the trend of k_m as function of p_j reported in Figure 10a. The latter plots the trend of Equation (6) for B-series and C-series assuming $\eta = 0.318$. Moreover, experimental results are represented for both series in correspondence of the average service pressure, i.e., 4 bar for low pressure and 36 for high pressure, and similarly is done for experimental results of Łątka and Matysek (2017), belonging to flat jack of B-series. Additionally, the analytical expression proposed by Zubkhov et al., which was reported in Łątka and Matysek (2017), is represented for the sake of comparison. It is described by the following formula:

$$k_m = 0.172 \ln(p_j) + 0.3591 \quad (7)$$

with p_j expressed in bar.

Furthermore, experimental results in terms of k_m are also plotted in Figure 10b for each specimen and for each pressure value, e.g., from 2–6 bar for low pressure and from 12–60 bar for high pressure, in order to have a comparison in a broader range of p_j and assessing the reliability of the model. In this case, values of k_m were obtained as the slope of load cycle curves for each pressure increment.

It is clear to observe as proposed expression allows to predict the average conversion factor in both ranges of pressure. In particular, if average pressures are considered (Figure 10a), Equation (6) gives for B-series

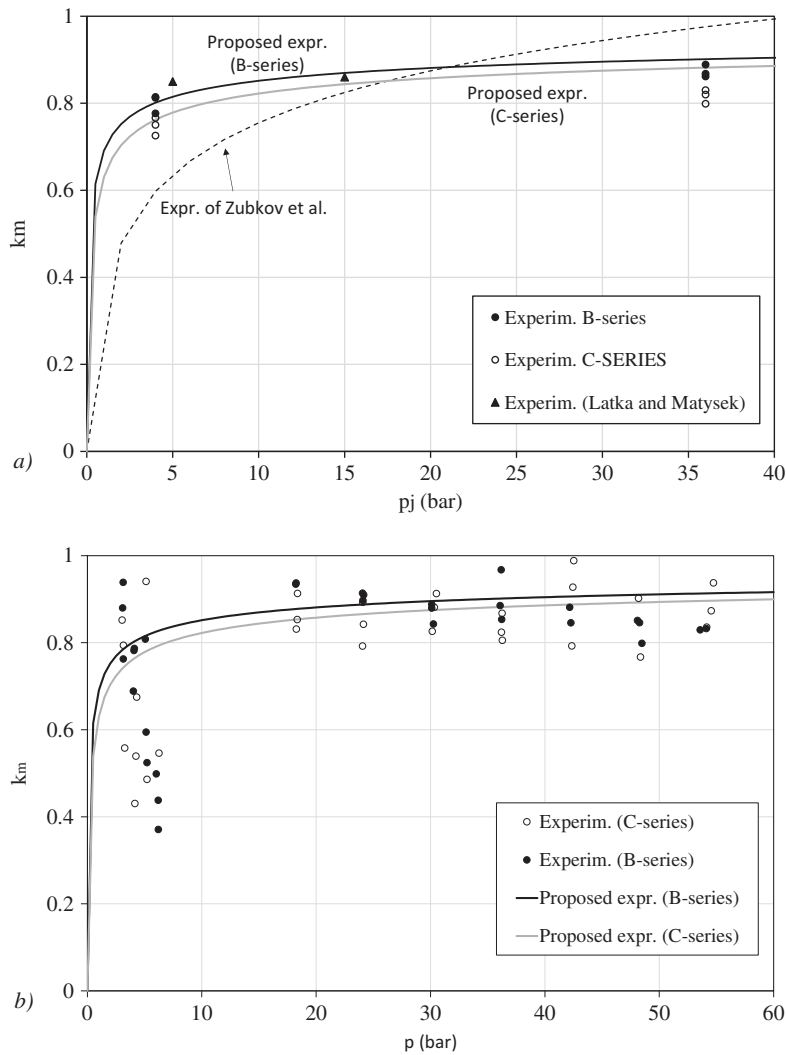


Figure 10. Comparison between proposed expression for k_m and experimental results: (a) comparisons with average values in the working pressure range; (b) comparisons with values deduced for each pressure increase.

$k_m = 0.80$ for low pressure and $k_m = 0.87$ for high pressure, values in accordance with recorded experimental data. Similarly, proposed expression provides estimations of conversion factor for C-series equal to $k_m = 0.76$ for low pressure and $k_m = 0.84$ for high pressure, with differences from experimental values equal to 1% and 3%, respectively. A straightforward comparison can be made by observing values reported in Table 1, which summarizes values of conversion factor deduced experimentally for each specimen $k_{m,exp}$ and values predicted theoretically $k_{m,th}$. It is evident as the proposed approach provides averagely reliable predictions for the range of pressure analyzed.

It is also stressed as experimental values of Łatka and Matysek (2017) for jack type B seems to follow the trend of Equation (6) for intermediate pressure values. Predicted values for these are $k_m = 0.81$ for $p_j = 5$ bar and $k_m = 0.87$ for $p_j = 15$ bar, with differences from

Table 1. Comparison between experimental and predicted values of k_m .

Specimen	p_j [bar] (Average)	$k_{m,exp}$	$k_{m,exp}$ (Average)	$k_{m,th}$	$k_{m,th}/k_{m,exp}$
B1	4	0.81	0.80	0.80	1.00
B2	4	0.78			
B3	4	0.81			
B1	36	0.89	0.87	0.87	1.00
B2	36	0.86			
B3	36	0.87			
C1	4	0.77	0.75	0.76	1.02
C2	4	0.73			
C3	4	0.75			
C1	36	0.83	0.82	0.84	1.03
C2	36	0.82			
C3	36	0.80			

experimental values equal to 5% and 1%, respectively. Moreover, Simoes et al. (2016) calibrated a flat jack with similar geometrical characteristics of C-series, even if the thickness was not provided, and obtained a

value of $k_m = 0.76$, which is in accordance with current obtained experimental results and theoretical predictions (Table 1).

Equation (7) gives value of k_m over 1 for $p_j > 40$ bar and it provides underestimations of the k_m coefficient for low pressures and overestimations for high pressures, as observed also in Gregorczyk and Lourenco (2000).

However, results plotted in Figure 10b show as the result is significantly dependent by the pressure range. In fact, it is evident as the value of k_m tends to a constant value for increasing values of pressure ($p_j > 12$ bar), while the dispersion of results obtained for each cycle is quite large for lower values of pressure. This consideration suggests again that experimental calibration of k_m should be performed for small pressure cycles, especially when low values of measured stress are expected, since the assessment of an average value of k_m can be significantly different from the actual value.

However, proposed model Equation (6) allows a simplified estimation of the conversion factor for high pressure values, once p_0 is calculated. In absence of a proper experimental calibration, it allows estimating the calibration curve of a flat jack, to be used for filtering results from in-situ testing. Differently, if experimental calibration is made for a broad band of stresses—which can be done by producers—the constants of Equation (6) can be estimated and compared to theoretical results, providing an overall calibration curve instead of only a predefined value of k_m . However, further comparisons are needed for assessing the reliability of proposed approach, which seems to be suitable for tested flat jacks.

5. Conclusions

This article presented the results of an experimental investigation on the calibration of k_m coefficient for the flat jack testing technique. An experimental investigation was performed, by calibrating two kinds of flat jack, produced by different manufacturers. On the basis of results achieved and for the range of investigated variables, the following conclusions can be drawn:

- the reliability of the flat-jack method is strictly dependent by the value of pressure induced, consequently by the stress measured in-situ for the single flat-jack technique, or by the mechanical properties of investigated masonry for double flat-jacks method;
- generally, more accurate results are expected for higher values of stress. The absence of a proper

calibration could lead to large errors, considering the sensitivity of k_m and A_{eff} from key parameters. Performed calibration allows reducing the scattering of the k_m coefficient to 4% for low pressure and 3% for high pressure, confirming the need for a preliminary calibration between performing a test; and

- a simplified method is proposed to assess the calibration curve of a flat jack, i.e., obtaining the conversion factor as a function of the working pressure. The proposed expression provides good estimations of the experimental data for an average range of the pressure in the jack. It could be useful for simplified estimations of the conversion factor but more comparisons with experimental investigations need to be addressed for a larger validation.

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